

ABDUL WAHAB BASIT

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EDUCATION

University of Mines and Technology (UMaT) | Tarkwa, Ghana

Sep 2019 – Sep 2023

Bachelor of Science, Computer Science & Engineering

Second Class Upper Honors

Relevant Coursework: Artificial Intelligence, Signal & Image Processing, Embedded Systems, Data Structures & Algorithms, Database Systems

RESEARCH INTERESTS

Machine Learning, Health Informatics, Responsible AI, Natural Language Processing, AI for Public and Population Health, Clinical Decision Support Systems, Data Visualization, Trustworthy AI, Scalable AI Systems

RESEARCH EXPERIENCE

Research Assistant, University of Mines and Technology

Tarkwa, Ghana

Advisor: Associate Professor Emmanuel Daanoba Sunkari

Jan 2024 – Mar 2025

Groundwater Quality Prediction and Health Risk Assessment Using Machine Learning

- Developed machine learning regression models including XGBoost, SVM, and Stochastic Gradient Descent to predict fluoride and nitrate contamination levels associated with public health risks in groundwater systems
- Applied Principal Component Analysis (PCA) and hydrogeochemical analysis to identify contamination drivers affecting vulnerable populations in Northern Ghana
- Evaluated predictive performance using RMSE and MSE metrics to support data-driven environmental and population health assessment.

Hypothyroid Disease Prediction Using Machine Learning

- Developed predictive classification models using Artificial Neural Networks (ANN), Random Forest, SVM, and Logistic Regression for hypothyroidism diagnosis, improving diagnostic AUROC from 0.73 to 0.89 through domain-informed feature engineering
- Conducted preprocessing and optimization of healthcare datasets to improve predictive accuracy and diagnostic reliability
- Compared model performance using precision, recall, and accuracy metrics, with ANN demonstrating the strongest predictive capability

Machine Learning for Rock Type Prediction

- Developed supervised learning models including Random Forest, Gradient Boosting, AdaBoost, and SVM for lithological classification using geochemical datasets from the Obuasi Gold District, Ghana
- Performed feature engineering, preprocessing, and comparative evaluation of predictive models to improve geological prediction accuracy
- Contributed to publication preparation and scientific reporting for peer-reviewed dissemination

PROFESSIONAL EXPERIENCE

Data Engineer

AmaliTech

East Legon, Accra, Ghana

Mar 2025 – Present

- Designed scalable ETL and ELT pipelines using PySpark and Databricks for high-volume streaming and batch-processing workloads
- Processed millions of records from APIs and Amazon S3 sources within AWS Lakehouse environments
- Improved ingestion efficiency by 70% through optimization of IoT and eCommerce data streaming architectures
- Automated data workflows to reduce manual intervention and support real-time analytics systems
- Worked with distributed systems technologies including Apache Spark, Kafka, AWS, and Microsoft Fabric

Data Annotator

Accra, Ghana

Ayadata.ai

Nov 2024 – Feb 2025

- Annotated and validated 10,000+ multimodal datasets including text, image, and audio data using CVAT and Labelbox
- Identified anomalies and edge cases to improve data quality for computer vision and NLP model training workflows
- Maintained annotation accuracy exceeding 95% across multiple machine learning projects

SELECTED RESEARCH PROJECTS

AI-Enabled Diabetes Screening Platform | *Project Lead*

- Leading a multidisciplinary team developing an AI-driven mobile screening platform for community healthcare settings in Ghana to facilitate deployable risk scoring without specialist oversight
- Designing automated machine learning and health analytics workflows to deliver large-scale risk assessment and early preventive healthcare intervention
- Optimizing client-side inference models to ensure system reliability and low-latency operation in infrastructurally constrained, low-resource environments

Speech-Therapy Application for Aphasic Patients | *Lead Developer (Undergraduate Capstone)*

- Designed and deployed a full-stack web application providing personalized language support, interactive speech therapy exercises, and contextual rehabilitation modules for aphasic patients
- Implemented NLP-driven response validation algorithms to dynamically assess user speech patterns, and track cognitive rehabilitation progress
- Evaluated system adaptability boundaries under evolving patient needs, directly informing research tracks into robust, linguistically diverse, and contextual health-AI technologies

Machine Learning-Based Sentiment Analysis in Combating Misinformation | *Independent Researcher*

- Implemented machine learning algorithms including SVM, Naive Bayes, and Neural Networks for sentiment classification of misinformation-related social media content
- Investigated hybrid modeling approaches and prompt engineering workflows to improve misinformation detection accuracy and reduce false positives across text datasets
- Explored NLP-based workflows for analyzing public sentiment and online information reliability

PUBLICATIONS

Peer-Reviewed & Journal Publications

- Sunkari, E. D., Muhammed, A. B., & **Basit, A. W.** (2025). *Application of Machine Learning Algorithms to Predict Rock Types Using Geochemical Data: A Case Study from the Obuasi Gold District, Ghana*. In E. D. Sunkari & S. Ali (Eds.), *Geology and Gold Mineralization in Ghana*. Earth and Environmental Sciences Library. Springer, Cham.
- Sunkari, E. D., Yvonne, O. B., Dickson, A. W., Moses, B. O., **Basit, A. W.**, & Michael, A. T. (2025). *Groundwater Fluoride and Nitrate Contamination in Gushegu Municipality, Northern Ghana: Hydrogeochemical Drivers, Machine-Learning Prediction, and Human Health Risks*. *Journal of African Earth*

Manuscripts in Progress

- ♦ Affum, E., **Basit, A. W.**, et al. (2024). *Advancing Network Intrusion Detection Leveraging Deep Convolutional Neural Networks*. Under Revision.
- ♦ Affum, E., **Basit, A. W.**, et al. (2024). *Advancing Hypothyroid Disease Prediction Using Machine Learning*. Under Revision.

Conference Abstracts & Presentations

- ♦ **Basit, A. W.**, et al. (2024). *A review on machine learning-based sentiment analysis in combating misinformation: techniques, challenges, and applications*. VI. International Turkic World Congress on Science and Engineering, 2024, Baku, Azerbaijan.

SKILLS

- ♦ **Programming:** Python, SQL, JavaScript, R
- ♦ **Machine Learning & AI:** TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, ANN, SVM, NLP Workflows, Prompt Engineering, LLM Experimentation
- ♦ **Data Engineering & Big Data:** PySpark, Apache Spark, Apache Kafka, Databricks, ETL/ELT Pipelines
- ♦ **Cloud & Infrastructure:** AWS, Microsoft Fabric, Docker, Git, Linux
- ♦ **Research & Analytics:** Causal Inference & Statistical Modeling, Classification, Regression, PCA, Data Preprocessing, Feature Engineering, Model Evaluation
- ♦ **Visualization:** Power BI, Matplotlib, Tableau

LEADERSHIP & VOLUNTEERING EXPERIENCE

Undergraduate Research Mentor

University of Mines and Technology

- ♦ Mentored undergraduate students in experimental design, machine learning implementation, and Python/R-based data analysis
- ♦ Guided students on research methodology and technical project development

STEM Education Volunteer

Rural Communities, Ghana

- ♦ Delivered STEM and technology awareness workshops across multiple rural communities in Ghana
- ♦ Mentored secondary school students on opportunities in AI, computing, and engineering

HONORS AND AWARDS

- ♦ Research Recognition Award | Process and Plant Automation Limited | Oct 2023
- ♦ Project Excellence Award | University of Mines and Technology | Aug 2022
- ♦ Industrial Training Slot Award | AngloGold Ashanti Iduapriem Limited | Jun 2021

NON-DEGREE CERTIFICATIONS

- ♦ AWS Certified Data Engineer – Associate, Amazon Web Services, 2025
- ♦ *Diploma in Using Python for Data Science*, Alison, 2024
- ♦ *Unsupervised & Supervised Machine Learning (Regression & Classification)*, Coursera, 2024